• a PCV valve attached to the left/front side of the valve cover (Fig. 2). It is secured with 1 bolt. An o-ring is used to seal valve to valve cover (Fig. 3). Another type of threaded PCV valve may be used (Fig. 4).
• the air cleaner housing
• tubes and hoses to connect the system components.

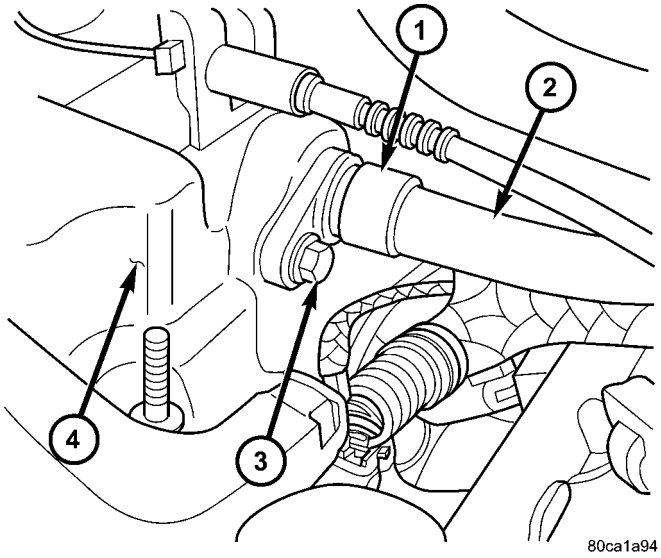


Fig. 2 PCV VALVE LOCATION - 2.4L

1 - PCV VALVE
2 - HOSE
3 - MOUNTING BOLT
4 - VALVE COVER (LEFT SIDE)

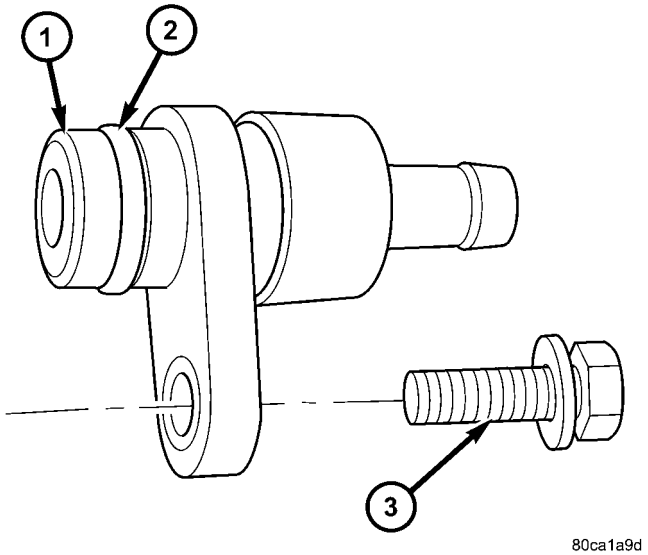


Fig. 3 PCV VALVE AND O-RING - 2.4L

1 - PCV VALVE
2 - O-RING
3 - MOUNTING BOLT

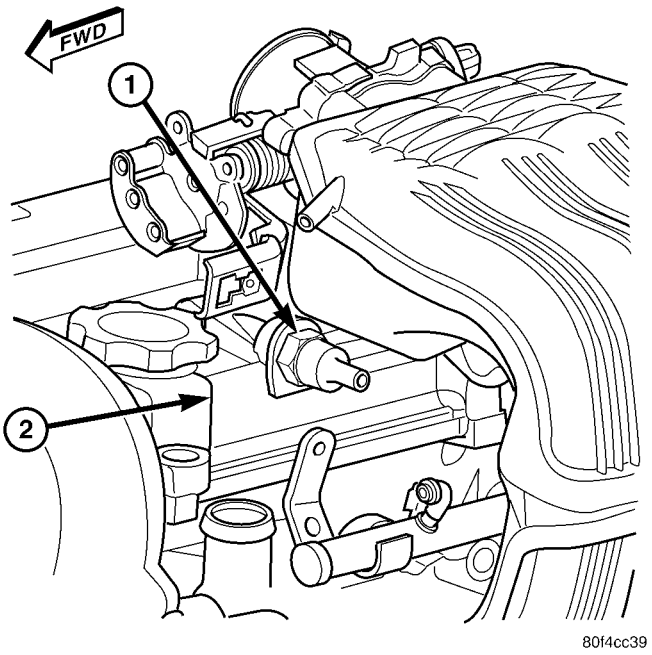


Fig. 4 PCV VALVE - 2.4L - THREADED

1 - PCV VALVE (THREADED)
2 - VALVE COVER (LEFT SIDE)

OPERATION

2.4L

The PCV system operates by engine intake manifold vacuum. Filtered air is routed into the crankcase through the air cleaner hose and crankcase breather(s) (if used). The metered air, along with crankcase vapors, are drawn through the PCV valve and into a passage in the intake manifold. The PCV system manages crankcase pressure and meters blow-by gases to the intake system, reducing engine sludge formation.

The PCV valve contains a spring loaded plunger. This plunger meters the amount of crankcase vapors routed into the combustion chamber based on intake manifold vacuum.

TYPICAL PCV valves are shown in (Fig. 5), (Fig. 6) and (Fig. 7).

When the engine is not operating, or during an engine pop-back, the spring forces the plunger back against the seat (Fig. 5). This will prevent vapors from flowing through the valve.

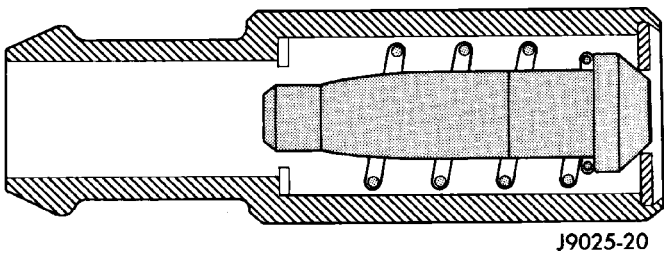


Fig. 5 ENGINE OFF OR ENGINE BACKFIRE - NO VAPOR FLOW